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PRESSURIZING MECHANISM, PRESSURIZING APPARATUS, SILO APPARATUS, AND BATTERY PRODUCTION LINE

Abstract

A pressurizing mechanism, a pressurizing apparatus, a silo apparatus, and a battery production line are disclosed. The pressurizing mechanism includes a base and a pressurizing piece. The base is provided with a positioning key. The pressurizing piece is provided with a positioning groove. The positioning key is fitted to the positioning groove to position the pressurizing piece and the base. The positioning groove is provided with an insertion opening running through a bottom wall of the pressurizing piece, and the positioning key is inserted into the positioning groove through the insertion opening.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International application PCT/CN2023/134688 filed on Nov. 28, 2023 that claims priority to the Chinese patent application No. 202311310323.6, filed on Oct. 11, 2023. The content of these applications is incorporated herein by reference in their entirety.

TECHNICAL FIELD
[0002] This disclosure relates to the technical field of batteries, and in particular, to a pressurizing mechanism, a pressurizing apparatus, a silo apparatus, and a battery production line.
BACKGROUND
[0003] This section is intended to provide background or context for the implementations disclosed in this disclosure. The description herein is not admitted to be prior art just because it is included in this section.
[0004] New energy batteries are increasingly widely used in life and industry. For example, new energy vehicles equipped with batteries have been widely used. In addition, batteries are increasingly used in the field of energy storage, and the like.
[0005] Taking a traction battery pack as an example, the traction battery pack includes a plurality of battery modules, and the battery modules need to be pressurized during the assembly process. To accurately and efficiently pressurize battery modules with different sizes, pressurizing pieces of pressurizing apparatus need to be adapted to the sizes of the battery modules, so it is necessary to use different pressurizing pieces to pressurize different battery modules. However, the replacement operation of pressurizing pieces on the existing production line is complicated and the replacement efficiency is low.
SUMMARY
[0006] To solve the above technical problems, this disclosure provides a pressurizing mechanism, a pressurizing apparatus, a silo apparatus, and a battery production line, which can solve the problems of complicated replacement operation and low replacement efficiency of pressurizing pieces.
[0007] This disclosure is implemented through the following technical solutions.
[0008] A first aspect of this disclosure provides a pressurizing mechanism. The pressurizing mechanism includes: [0009] a base; and [0010] a pressurizing piece arranged on the base, where the base is provided with a positioning key, the pressurizing piece is provided with a positioning groove, and the positioning key is fitted to the positioning groove to position the pressurizing piece and the base; and [0011] the positioning groove is provided with an insertion opening running through a bottom wall of the pressurizing piece, and the positioning key is inserted into the positioning groove through the insertion opening.
[0012] The positioning key is arranged on the base, and the pressurizing piece is provided with the positioning groove, so that the pressurizing piece and the base can be positioned through the fitting of the positioning key and the positioning groove. The pressurizing mechanism is simple and convenient to operate, and facilitates the connection and disconnection between the pressurizing

piece and the base, thereby alleviating the problems of complicated replacement operation and low replacement efficiency of the pressurizing piece. In addition, the pressurizing piece and the base are positioned through the fitting of the positioning key and the positioning groove, which is conducive the automation to some extent. Further, different types of pressurizing pieces can be provided with positioning keys or positioning grooves adapted to the base, so that different types of pressurizing pieces can share the base, thereby reducing the cost and improving the production efficiency. In addition, with the insertion opening formed in the bottom wall of the pressurizing piece, the positioning key can be inserted into the positioning groove through the insertion opening at the bottom of the pressurizing piece, so that the pressurizing piece can be moved in the height direction and assembled to the base. The operation is simple and reliable, which is conducive to automation.
[0013] In some embodiments, the base is provided with an avoidance groove, and the avoidance groove is configured to avoid the pressurizing piece.
[0014] The avoidance groove is provided on the base to avoid a protruding portion of the pressurizing piece, so that the pressurizing piece can be more firmly fixed on the base. In addition, the versatility of the base can be improved to adapt to different types of pressurizing pieces.
[0015] In some embodiments, the pressurizing piece includes a connecting portion and a pressurizing portion connected to the connecting portion, and the pressurizing portion is provided with the positioning groove.
[0016] The pressurizing piece is arranged in a discrete structure including the connecting portion and the pressurizing portion, and is connected to the base through the connecting portion, and comes into contact with a battery module through the pressurizing portion. Therefore, the strength of the connecting portion can be ensured, and the strength of a connecting structure between the pressurizing piece and the base can be improved. In addition, the contact between the pressurizing portion and the battery module can prevent the pressurizing portion from damaging the battery module to some extent. Moreover, the molding and manufacturing are simple.
[0017] In some embodiments, the pressurizing portion includes a connecting body and a limiting piece, where the connecting body is provided with a groove, and the limiting piece is arranged at an edge of the groove and defines the positioning groove with the groove.
[0018] The pressurizing portion includes the connecting body and the limiting piece, and the limiting piece is disposed at the edge of the groove and defines the positioning groove with the groove. This is conducive to shaping the required positioning groove on the premise of ensuring the strength of the pressurizing portion, so as to realize the connection between the pressurizing piece and the base.
[0019] In some embodiments, the positioning groove is provided with an insertion opening running through a bottom wall of the pressurizing piece, and the positioning key is inserted into the positioning groove through the insertion opening.
[0020] With the insertion opening formed in the bottom wall of the pressurizing piece, the positioning key can be inserted into the positioning groove through the insertion opening at the bottom of the pressurizing piece, so that the pressurizing piece can be moved in the height direction and assembled to the base. The operation is simple and reliable, which is conducive to automation.
[0021] In some embodiments, the pressurizing piece is provided with a grip portion at the top, and the grip portion is configured to cooperate with a manipulator in gripping.
[0022] The provision of the grip portion arranged at the top of the pressurizing piece is more conducive to operating the pressurizing piece, thereby further alleviating the problems of complicated replacement operation and low replacement efficiency of the pressurizing piece. The grip portion is configured to cooperate with the manipulator in gripping, which is conducive to the automation of pressurizing piece replacement.
[0023] In some embodiments, the base is provided with a plurality of positioning keys.
[0024] The plurality of positioning keys can be arranged on the base, that is, the base can position the plurality of pressurizing pieces simultaneously, fully utilizing the space of the base.

[0025] In some embodiments, the base includes a base body and a fastener arranged on the base body, and the positioning key is arranged on a side surface of the fastener.
[0026] With the fastener provided and the positioning key arranged on the side surface of the fastener, the height of the fastener can be adjusted according to the height of the pressurizing piece, or the height of the positioning key can be adjusted according to the height of the positioning groove, supporting adaptation to more pressurizing pieces. In addition, when the positioning key is inserted into the positioning groove, the side surface of the pressurizing piece can also be attached to the side surface of the fastener, and the bottom surface of the pressurizing piece can also be attached to the surface of the base body, thereby improving the reliability of the connection structure between the pressurizing piece and the base.
[0027] In some embodiments, the pressurizing piece includes a connecting body and a limiting piece, where the connecting body is provided with a groove, and the limiting piece is arranged at an edge of the groove and defines the positioning groove with the groove.
[0028] The positioning key is arranged on a side surface of the fastener, and a limiting groove is formed between the positioning key and the fastener; and when the positioning key is fitted to the positioning groove, the limiting piece is snapped into the limiting groove to limit the pressurizing piece and the base.
[0029] With the limiting groove formed between the positioning key and the fastener, the limiting piece is snapped into the limiting groove when the positioning key is fitted to the positioning groove, so as to limit the pressurizing piece and the base. This further improves the connection reliability between the pressurizing piece and the base.
[0030] In some embodiments, the fastener is provided with a fool-proofing groove at the top, and the pressurizing piece is provided with a fool-proofing piece in one-to-one correspondence with the fool-proofing groove at the top, and when the positioning key is fitted to the positioning groove, the fool-proofing piece is snapped into the fool-proofing groove, where dimensions of the fool-proofing pieces of different pressurizing pieces are different.
[0031] The fastener is provided with the fool-proofing groove at the top, and the pressurizing piece is provided with the fool-proofing piece in one-to-one correspondence with the fool-proofing groove at the top, so that the accuracy of replacing the pressurizing piece can be ensured, and the wrong pressurizing piece can be prevented from damaging the battery module. In addition, the fool-proofing piece is fitted to the fool-proofing groove, which can further position the pressurizing piece and the base.
[0032] In some embodiments, a width of the base is L**1**, where 100 mm≤L**1**≤1200 mm.
[0033] This can achieve compatibility with more pressurizing pieces, as well as more single-row or double-row pressurizing pieces of more projects, and can also make the structure of a pressurizing apparatus or a silo apparatus compact.
[0034] In some embodiments, a width of the base is L**1**, where 150 mm≤L**1**≤500 mm.
[0035] This can achieve compatibility with more pressurizing pieces, as well as more single-row or double-row pressurizing pieces of more projects, and can further make the structure of the pressurizing apparatus or the silo apparatus more compact.
[0036] A second aspect of this disclosure provides a pressurizing apparatus. The pressurizing apparatus includes a tray and the pressurizing mechanism disclosed in any one of the above embodiments. The pressurizing mechanism is arranged on the tray, and the pressurizing mechanism is configured to pressurize a battery module on the tray.
[0037] The positioning key is arranged on the base, and the pressurizing piece is provided with the positioning groove, so that the pressurizing piece and the base can be positioned through the fitting of the positioning key and the positioning groove. The pressurizing mechanism is simple and convenient to operate, and facilitates the connection and disconnection between the pressurizing piece and the base, thereby alleviating the problems of complicated replacement operation and low replacement efficiency of the pressurizing piece. In addition, the pressurizing piece and the base are

positioned through the fitting of the positioning key and the positioning groove, which is conducive the automation to some extent. Further, different types of pressurizing pieces can be provided with positioning keys or positioning grooves adapted to the base, so that different types of pressurizing pieces can share the base, thereby reducing the cost and improving the production efficiency. In addition, with the insertion opening formed in the bottom wall of the pressurizing piece, the positioning key can be inserted into the positioning groove through the insertion opening at the bottom of the pressurizing piece, so that the pressurizing piece can be moved in the height direction and assembled to the base. The operation is simple and reliable, which is conducive to automation.
[0038] A third aspect of this disclosure provides a silo apparatus. The silo apparatus includes a case and the pressurizing mechanism disclosed in any one of the above embodiments. The case is provided with a feeding area and an unloading area. The feeding area and the unloading area are both provided with the pressurizing mechanism.
[0039] The positioning key is arranged on the base, and the pressurizing piece is provided with the positioning groove, so that the pressurizing piece and the base can be positioned through the fitting of the positioning key and the positioning groove. The pressurizing mechanism is simple and convenient to operate, and facilitates the connection and disconnection between the pressurizing piece and the base, thereby alleviating the problems of complicated replacement operation and low replacement efficiency of the pressurizing piece. In addition, the pressurizing piece and the base are positioned through the fitting of the positioning key and the positioning groove, which is conducive the automation to some extent. Further, different types of pressurizing pieces can be provided with positioning keys or positioning grooves adapted to the base, so that different types of pressurizing pieces can share the base, thereby reducing the cost and improving the production efficiency. In addition, with the insertion opening formed in the bottom wall of the pressurizing piece, the positioning key can be inserted into the positioning groove through the insertion opening at the bottom of the pressurizing piece, so that the pressurizing piece can be moved in the height direction and assembled to the base. The operation is simple and reliable, which is conducive to automation.
[0040] In some embodiments, the feeding area includes a plurality of feeding sub-areas arranged along a height direction, and each of the feeding sub-areas is provided with a plurality of pressurizing mechanisms.
[0041] The feeding area is designed to include a plurality of feeding sub-areas arranged along the height direction, so that the space of the case can be fully utilized, the structure is compact, and more pressurizing mechanisms can be placed in the feeding area, improving the production efficiency.
[0042] In some embodiments, the unloading area includes a plurality of unloading sub-areas arranged along the height direction, and each of the unloading sub-areas is provided with a plurality of pressurizing mechanisms.
[0043] The unloading area is designed to include a plurality of unloading sub-areas arranged along the height direction, so that the space of the case can be fully utilized, the structure is compact, and more pressurizing mechanisms can be placed in the unloading area, improving the production efficiency.
[0044] A fourth aspect of this disclosure provides a battery production line including a manipulator, the pressurizing apparatus disclosed in any one of the above embodiments, and the silo apparatus disclosed in any one of the above embodiments. The manipulator is configured to transfer and assemble the pressurizing piece in the feeding area to the base of the pressurizing apparatus, and transfer and assemble the to-be-replaced pressurizing piece in the pressurizing apparatus to the base in the unloading area.
[0045] In this way, the movement of the manipulator is electrically controlled. Through the pressurizing mechanism and electrical control, the pressurizing piece can be automatically replaced, reducing the labor cost, improving the replacement efficiency and the production capacity

of the production line, and realizing replacement of the pressurizing piece without shutting down the machine.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0046] A person of ordinary skill in the art becomes clearly aware of various other advantages and benefits by reading the following detailed description of exemplary embodiments. The drawings are merely intended to illustrate the preferred implementations, but not to limit this disclosure. Moreover, throughout the accompanying drawings, the same reference signs represent the same parts. In the accompanying drawings:

[0047] FIG. **1** is a structural schematic diagram of a battery production line from a first perspective according to some embodiments of this disclosure;

[0048] FIG. **2** is a schematic structural view of a battery production line from a second perspective according to some embodiments of this disclosure;

[0049] FIG. **3** is a structural schematic diagram of a pressurizing mechanism according to some embodiments of this disclosure;

[0050] FIG. **4** is a structural schematic diagram of a pressurizing piece according to some embodiments of this disclosure;

[0051] FIG. **5** is an exploded view of a pressurizing piece and a base from a third perspective according to some other embodiments of this disclosure; and

[0052] FIG. **6** is an exploded view of a pressurizing piece and a base from a fourth perspective according to some other embodiments of this disclosure.

REFERENCE SIGNS

[0053] **10**. base; **10***a*. positioning key; **10***b*. avoidance groove; **10***c*. limiting groove; **11**. base body; **12**. fastener; **20**. pressurizing piece; **20***a*. positioning groove; **20***b*. grip portion; **20***c*. fool-proofing piece; **21**. pressurizing portion; **211**. connecting body; **212**. limiting piece; **22**. connecting portion; **100**. pressurizing apparatus; **200**. silo apparatus; **210**. case; **210***a*. feeding area; **210***b*. unloading area; **220**. storage vehicle; **300**. manipulator; and **1000**. battery production line.

DETAILED DESCRIPTION

[0054] Embodiments of the technical solutions of this disclosure are described in detail below with reference to the drawings. The following embodiments are merely intended to describe the technical solutions of this disclosure more clearly, and are merely exemplary but without hereby limiting the protection scope of this disclosure.

[0055] Unless otherwise defined, all technical and scientific terms used herein have the same meanings as normally understood by a person skilled in the technical field of this disclosure. The terms used herein are merely intended to describe specific embodiments and but not to limit this disclosure. The terms “include” and “contain” and any variations thereof in this disclosure are intended as non-exclusive inclusion.

[0056] In the description of the embodiments of this disclosure, the technical terms “first”, “second” and “third” are merely intended to distinguish between different items but not intended to indicate or imply relative importance or implicitly specify the quantity of the indicated technical features, specific order, or order of precedence. In the description of the embodiments of this disclosure, unless otherwise expressly specified, “a plurality of” means two or more.

[0057] Reference to “embodiment” herein means that a specific feature, structure or characteristic described with reference to the embodiment may be included in at least one embodiment of this disclosure. Reference to this term in different places in the specification does not necessarily represent the same embodiment, nor does it represent an independent or alternative embodiment in a mutually exclusive relationship with other embodiments. A person skilled in the art explicitly and

implicitly understands that the embodiments described herein may be combined with other embodiments.
[0058] In the description of embodiments of this disclosure, the term "and/or" merely indicates a relationship between related items, and represents three possible relationships. For example, "A and/or B" may represent the following three circumstances: A alone, both A and B, and B alone. In addition, the character "/" herein generally indicates an "or" relationship between contextually associated objects.
[0059] In the description of the embodiments of this disclosure, direction or a positional relationship indicated by the terms such as "length", "width", "thickness", "up", "down", "before", "after", "left", "right", "vertical", "horizontal", "top", "bottom", "in", "out" and "clockwise" is a direction or positional relationship based on the illustration in the drawings, and is merely intended for ease or brevity of description of embodiments of this disclosure, but not intended to indicate or imply that the indicated apparatus or component is necessarily located in the specified direction or constructed or operated or used in the specified direction. Therefore, such terms are not to be understood as a limitation on embodiments of this disclosure.
[0060] In the description of the embodiments of this disclosure, unless otherwise explicitly provided and limited, the technical terms such as "mount", "connect", "couple", and "fix" should be understood broadly, which, for example, may refer to a fixed connection, a detachable connection, or an integral connection; or may refer to a mechanical connection or an electrical connection; or may refer to a direct connection or an indirect connection via an intermediate medium; or may refer to a communication between the insides of two elements, or an interaction relationship between two elements. For those of ordinary skill in the art, the specific meanings of the above-mentioned terms in the embodiments of this disclosure may be understood according to specific situations.
[0061] In the description of the embodiments of this disclosure, unless otherwise explicitly provided and limited, the technical term "contact" should be broadly understood, which may be direct contact, contact through an intermediate medium layer, contact with no interaction force between the two in contact, or contact with interaction force between the two in contact.
[0062] The vigorous promotion of new energy vehicles by the authorities ushers in a great opportunity for the development of new energy vehicles. The safety and stability of vehicles have always been people's top concern. Therefore, improving the safety of new energy vehicles will be one of the important factors that determine whether new energy vehicles can be popularized quickly. As a main component of the battery pack of new energy vehicles, improving the safety of battery modules is an important approach to improving the safety of new energy vehicles.
[0063] A battery module usually includes a plurality of battery cells, the battery cells are confined in a module frame formed by welding end plates and side plates. In the assembly process of the module, the end plates at both ends of the module have a certain pre-tightening force on the battery to improve the cycling performance of the battery.
[0064] Therefore, in the process of assembling the battery module, it is necessary to pressurize the battery module so that the end plates at both ends of the module have a certain pre-tightening force on the battery. To accurately and efficiently pressurize battery modules with different sizes, pressurizing pieces of pressurizing apparatus need to be adapted to the sizes of the battery modules, so it is necessary to replace different pressurizing pieces to pressurize different battery modules.
[0065] However, the pressurizing pieces on the existing production line are usually fixed on a silo or a tray by fastening, which leads to complicated replacement operation and low replacement efficiency of the pressurizing pieces, and is not conducive to automation.
[0066] To solve the problems that the replacement operation of the pressurizing pieces is complicated, the replacement efficiency is low, and automation is hard to achieve, referring to FIGS. **1** to **6**, this disclosure designs a pressurizing mechanism. The pressurizing mechanism includes a base **10** and a pressurizing piece **20**. One of the pressurizing piece **20** and the base **10** is

provided with a positioning key **10***a*, and the other one is provided with a positioning groove **20***a*. The positioning key **10***a* is fitted to the positioning groove **20***a* to position the pressurizing piece **20** and the base **10**.

[0067] In this pressurizing mechanism, the positioning key **10***a* is arranged on the base **10**, and the positioning groove **20***a* is arranged on the pressurizing piece **20**, so that the pressurizing piece **20** and the base **10** can be positioned through the fitting of the positioning key **10***a* and the positioning groove **20***a*. This pressurizing mechanism is simple and convenient to operate, which is conducive to the connection and disconnection between the pressurizing piece **20** and the base **10**, thereby alleviating the problems of complicated replacement operation and low replacement efficiency of the pressurizing piece **20**. In addition, the pressurizing piece **20** and the base **10** are positioned through the fitting of the positioning key **10***a* and the positioning groove **20***a*, which is conducive to automation to some extent. Further, different types of pressurizing pieces **20** can be provided with positioning keys **10***a* or positioning grooves **20***a* adapted to the base **10**, so that different types of pressurizing pieces **20** can share the base **10**, thereby reducing the cost and improving the production efficiency. In addition, with the insertion opening formed in the bottom wall of the pressurizing piece **20**, the positioning key **10***a* can be inserted into the positioning groove **20***a* through the insertion opening at the bottom of the pressurizing piece **20**, so that the pressurizing piece **20** can be moved in the height direction and assembled to the base **10**. The operation is simple and reliable, which is conducive to automation.

[0068] The pressurizing mechanism of this disclosure can be used not only for a pressurizing apparatus **100**, but also for a silo apparatus **200**, without being specifically limited.

[0069] According to some embodiments of this disclosure, referring to FIGS. **1** to **6**, FIG. **3** is a structural schematic diagram of a pressurizing mechanism according to some embodiments of this disclosure, FIG. **4** is a structural schematic diagram of a pressurizing piece according to some embodiments of this disclosure, FIG. **5** is an exploded view of a pressurizing piece and a base from a third perspective according to some other embodiments of this disclosure, and FIG. **6** is an exploded view of a pressurizing piece and a base from a fourth perspective according to some other embodiments of this disclosure. This disclosure provides a pressurizing mechanism. The pressurizing mechanism includes a base **10** and a pressurizing piece **20**. The base **10** is provided with a positioning key **10***a*. The pressurizing piece **20** is provided with a positioning groove **20***a*. The positioning key **10***a* is fitted to the positioning groove **20***a* to position the pressurizing piece **20** and the base **10**.

[0070] The base **10** herein may be a mounting seat of a pressurizing tooling for carrying the pressurizing piece **20** or a silo for carrying the pressurizing piece **20**.

[0071] The pressurizing piece **20** refers to the pressurizing piece **20** for pressurizing the end plate of the battery module. In the pressurizing process, the pressurizing piece **20** are in contact with the end plate of the battery module, so as to prevent the battery module from being damaged by the squeezing apparatus. Different battery modules are provided with different force-receiving positions, so for different types of battery modules, different pressurizing pieces **20** can be replaced to pressurize the battery modules.

[0072] It should be noted that the base **10** may be provided with a positioning groove **20***a*, and the pressurizing piece **20** may be provided with a positioning key **10***a*.

[0073] Certainly, in other embodiments, the base **10** may be provided with a positioning key **10***a* and a positioning groove **20***a*, and the pressurizing piece **20** may be provided with a positioning key **10***a* and a positioning groove **20***a* corresponding to the base **10**.

[0074] The positioning key **10***a* is fitted to the positioning groove **20***a* to position the pressurizing piece **20** and the base **10**. The positioning and connection between the pressurizing piece **20** and the base **10** can be realized simply by inserting the positioning key **10***a* into the positioning groove **20***a*. In other words, there is no need to fasten the connection through fasteners, which is conducive to automation.

[0075] For example, the positioning key **10***a* and the positioning groove **20***a* are in anti-rotation fit with each other, that is, after positioning the pressurizing piece **20** and the base **10** through the positioning key **10***a* and the positioning groove **20***a*, the pressurizing piece **20** cannot rotate relative to the base **10**.
[0076] According to the pressurizing mechanism provided in the embodiment of this disclosure, the base **10** is provided with the positioning key **10***a*, and the pressurizing piece **20** is provided with the positioning groove **20***a*, so that the pressurizing piece **20** and the base **10** are positioned through the fitting of the positioning key **10***a* and the positioning groove **20***a*. The installation structure is simple, the operation is convenient, and the connection and disconnection between the pressurizing piece **20** and the base **10** are facilitated, thereby solving the problems of complicated replacement operation and low replacement efficiency of the pressurizing piece **20**. In addition, the pressurizing piece **20** and the base **10** are positioned through the fitting of the positioning key **10***a* and the positioning groove **20***a*, which is conducive to automation to some extent. Further, different types of pressurizing pieces **20** can be provided with positioning keys **10***a* or positioning grooves **20***a* adapted to the base **10**, so that different types of pressurizing pieces **20** can share the base **10**, thereby reducing the cost and improving the production efficiency. In addition, with the insertion opening formed in the bottom wall of the pressurizing piece **20**, the positioning key **10***a* can be inserted into the positioning groove **20***a* through the insertion opening at the bottom of the pressurizing piece **20**, so that the pressurizing piece **20** can be moved in the height direction and assembled to the base **10**. The operation is simple and reliable, which is conducive to automation.
[0077] In some embodiments, referring to FIGS. **3** and **5**, the base **10** is provided with an avoidance groove **10***b*, and the avoidance groove **10***b* is configured to avoid the pressurizing piece **20**.
[0078] It should be noted that the specific shape of the avoidance groove **10***b* is not limited here, as long as the avoidance groove **10***b* can be configured to avoid the pressurizing piece **20**. For example, the shape of the avoidance groove **10***b* is polygonal, circular, elliptical or irregular.
[0079] It should be noted that the size of the avoidance groove **10***b* is not limited here. In some embodiments, the avoidance groove **10***b* penetrates the base **10**, for example, when the base **10** is a silo. Certainly, in other embodiments, the base **10** may be recessed to form the avoidance groove **10***b*.
[0080] To adapt to different types of battery modules, some pressurizing pieces **20** will form special-shaped protrusions for better pressurizing the battery modules. In this embodiment, the avoidance groove **10***b* is provided on the base **10** to avoid the protruding portion of the pressurizing piece **20**, so that the pressurizing piece **20** can be more firmly fixed on the base **10**. In addition, the versatility of the base **10** can be improved to adapt to different types of pressurizing pieces **20**.
[0081] In some embodiments, referring to FIGS. **3** and **4**, the pressurizing piece **20** includes a connecting portion **22** and a pressurizing portion **21** connected to the connecting portion **22**. The pressurizing portion **21** is provided with a positioning groove **20***a*.
[0082] For example, the connecting portion **22** and the pressurizing portion **21** may be discrete structures. The discrete connecting portion **22** and the discrete pressurizing portion **21** are conducive to forming the required structure.
[0083] Certainly, the pressurizing piece **20** may also be of an integrated structure. The integrated pressurizing piece **20** can reduce the number of components, shorten the assembly time and improve the assembly efficiency.
[0084] The connecting portion **22** and the pressurizing portion **21** are discrete in this embodiment of this disclosure, so that they can be connected to the base **10** through the connecting portion **22** and come into contact with the battery module through the pressurizing portion **21**.
[0085] The connection manner between the connecting portion **22** and the pressurizing portion **21** is not limited herein. For example, the connecting portion **22** and the pressurizing portion **21** can be connected by magnetic attraction, snap connection, fastening connection or plug connection.
[0086] The fastening connection includes, but is not limited to, screw connection, bolt connection,

rivet connection, or the like.

[0087] The specific material of the connecting portion **22** is not limited herein, for example, it can be a metal piece, so that the strength of the connecting portion **22** can be ensured, and the strength of the connecting structure between the pressurizing piece **20** and the base **10** can be improved.

[0088] The specific material of the pressurizing portion **21** is not limited herein, for example, it can be a plastic piece or a rubber piece, so that the pressurizing portion **21** can be prevented from damaging the battery module to some extent. Moreover, the molding and manufacturing are simple.

[0089] The pressurizing piece **20** is arranged in a discrete structure including the connecting portion **22** and the pressurizing portion **21**, and is connected to the base **10** through the connecting portion **22** and comes into contact with the battery module through the pressurizing portion **21**. Therefore, the strength of the connecting portion **22** can be ensured, and the strength of the connecting structure between the pressurizing piece **20** and the base **10** can be improved. In addition, the contact between the pressurizing portion **21** and the battery module can prevent the pressurizing portion **21** from damaging the battery module to some extent. Moreover, the molding and manufacturing are simple.

[0090] In some embodiments, referring to FIGS. **3** and **4**, the pressurizing portion **21** includes a connecting body **211** and a limiting piece **212**. The connecting body **211** is provided with a groove. The limiting piece **212** is arranged at the edge of the groove and defines a positioning groove **20***a* with the groove.

[0091] For example, the connecting body **211** and the limiting piece **212** may be discrete. The discrete connecting body **211** and the limiting piece **212** are conducive to forming the required structure.

[0092] Certainly, the pressurizing portion **21** may be an integrated structure. The integrated pressurizing portion **21** can reduce the number of components, shorten the assembly time, and improve the assembly efficiency.

[0093] The connecting body **211** is provided with a groove, and the specific shape of the groove is not limited herein.

[0094] The pressurizing portion **21** is of a discrete structure including the connecting body **211** and the limiting piece **212**, and the limiting piece **212** is disposed at the edge of the groove and defines the positioning groove **20***a* with the groove, so it is conducive to shaping the required positioning groove **20***a* on the premise of ensuring the strength of the pressurizing portion **21**, so as to realize the connection between the pressurizing piece **20** and the base **10**.

[0095] In some embodiments, referring to FIGS. **3** and **4**, the positioning groove **20***a* has an insertion opening penetrating the bottom wall of the pressurizing piece **20**. The positioning key **10***a* is inserted into the positioning groove **20***a* through the insertion opening.

[0096] The positioning groove **20***a* is provided with an insertion opening running through a bottom wall of the pressurizing piece **20**, that is, the positioning groove **20***a* penetrates through the bottom wall of the pressurizing piece **20** to form the insertion opening, so that the positioning key **10***a* can be inserted into the positioning groove **20***a* through the insertion opening at the bottom of the pressurizing piece **20**.

[0097] With the insertion opening formed in the bottom wall of the pressurizing piece **20**, the positioning key **10***a* can be inserted into the positioning groove **20***a* through the insertion opening at the bottom of the pressurizing piece **20**, so that the pressurizing piece **20** can be moved in the height direction and assembled to the base **10**. The operation is simple and reliable, which is conducive to automation.

[0098] In some embodiments, referring to FIGS. **3** and **4**, the pressurizing piece **20** is provided with a grip portion **20***b* at the top. The grip portion **20***b* is configured to cooperate with the manipulator **300** in gripping.

[0099] The pressurizing piece **20** is provided with a grip portion **20***b* at the top. With the grip portion **20***b* arranged at the top of the pressurizing piece **20**, it is more conducive to operating the pressurizing piece **20**, thereby further alleviating the problems of complicated replacement operation and low replacement efficiency of the pressurizing piece **20**.

[0100] The grip portion **20***b* is configured to cooperate with the manipulator **300** in gripping, thereby facilitating the automation of the replacement of the pressurizing piece **20**. Further, the provision of the grip portion **20***b* can improve the reliability of the manipulator **300** in griping the pressurizing piece **20**.

[0101] The grip portion **20***b* may be arranged at the middle position of the top of the pressurizing piece **20**, at arranged both ends of the top of the pressurizing piece **20**, or directly gripped by the manipulator **300** at both ends of the top of the pressurizing piece **20**.

[0102] The grip portion **20***b* is not limited to a specific structure herein, and it may be a protrusion formed on the pressurizing piece **20** or a groove formed on the pressurizing piece **20**. This embodiment of this disclosure uses the grip portion **20***b* being a groove formed on the pressurizing piece **20** as an example.

[0103] For example, the grip portion **20***b* includes a limiting flange protrudingly formed on a wall of the groove and configured to limit the manipulator **300**, further improving the reliability of the manipulator **300** in griping the pressurizing piece **20**.

[0104] For example, the grip portion **20***b* is formed on the connecting portion **22**, so that the strength of the grip portion **20***b* can be ensured, thereby further improving the reliability of the manipulator **300** in griping the pressurizing piece **20**.

[0105] With the grip portion **20***b* arranged at the top of the pressurizing piece **20**, it is more conducive to operating the pressurizing piece **20**, thereby further alleviating the problems of complicated replacement operation and low replacement efficiency of the pressurizing piece **20**. The grip portion **20***b* is configured to cooperate with the manipulator **300** in gripping, thereby facilitating the automation of the replacement of the pressurizing piece **20**.

[0106] In some embodiments, referring to FIG. **3**, the base **10** is provided with a plurality of positioning keys **10***a*, for example, 2, 3, 4, 5, 6, 7, 8, 9, or the like.

[0107] A plurality of positioning keys **10***a* are disposed on the base **10**, that is, the base **10** can position a plurality of pressurizing pieces **20** simultaneously, fully utilizing the space of the base **10**.

[0108] In a specific embodiment, the base **10** is provided with four positioning keys **10***a*.

[0109] In some embodiments, referring to FIGS. **3** to **6**, the base **10** includes a base body **11** and a fastener **12** disposed on the base body **11**. The positioning key **10***a* is provided on a side surface of the fastener **12**.

[0110] For example, the base body **11** and the fastener **12** may be discrete structures. The discrete base body **11** and fastener **12** are conducive to forming the required structure.

[0111] Certainly, the base **10** can also be an integrated structure. The integrated base **10** can reduce the number of components, shorten the assembly time, and improve the assembly efficiency.

[0112] With the fastener **12** provided and the positioning key **10***a* arranged on the side surface of the fastener **12**, the height of the fastener **12** can be adjusted according to the height of the pressurizing piece **20**, or the height of the positioning key **10***a* can be adjusted according to the height of the positioning groove **20***a*, supporting adaptation to more pressurizing pieces **20**. In addition, when the positioning key **10***a* is inserted into the positioning groove **20***a*, the side surface of the pressurizing piece **20** can also be attached to the side surface of the fastener **12**, and the bottom surface of the pressurizing piece **20** can also be attached to the surface of the base body **11**, thereby improving the reliability of the connection structure between the pressurizing piece **20** and the base **10**.

[0113] In some embodiments, referring to FIGS. **3** and **4**, the pressurizing piece **20** includes a connecting body **211** and a limiting piece **212**. The connecting body **211** is provided with a groove. The limiting piece **212** is arranged at the edge of the groove and defines a positioning groove **20***a* with the groove. The positioning key **10***a* is arranged on the side surface of the fastener **12**, and a limiting groove **10***c* is formed between the positioning key **10***a* and the fastener **12**. When the positioning key **10***a* is fitted to the positioning groove **20***a*, the limiting piece **212** is fitted into the limiting groove **10***c* to limit the pressurizing piece **20** and the base **10**.

[0114] The positioning key **10***a* is arranged on the side surface of the fastener **12**, and a limiting groove **10***c* is formed between the positioning key **10***a* and the fastener **12**. The specific type of the limiting groove **10***c* is not limited herein, for example, a part of an area of the positioning key **10***a* near one end of the fastener **12** is recessed inward, and the concave area and the side surface of the fastener **12** jointly define the limiting groove **10***c*.

[0115] The limiting groove **10***c* may surround the peripheral outer surface of the positioning key **10***a*, or it may partially surround the peripheral outer surface of the positioning key **10***a*.

[0116] With the limiting groove **10***c* formed between the positioning key **10***a* and the fastener **12**, the limiting piece **212** is snapped into the limiting groove **10***c* when the positioning key **10***a* is fitted to the positioning groove **20***a*, so as to limit the pressurizing piece **20** and the base **10**. This further improves the connection reliability between the pressurizing piece **20** and the base **10**.

[0117] In some embodiments, referring to FIGS. **5** and **6**, the fastener **12** is provided with a fool-proofing groove at the top. The pressurizing piece **20** is provided with a fool-proofing piece **20***c* in one-to-one correspondence with the fool-proofing groove at the top, and when the positioning key **10***a* is fitted to the positioning groove **20***a*, the fool-proofing piece **20***c* is snapped into the fool-proofing groove. The fool-proofing pieces **20***c* of different pressurizing pieces **20** are different.

[0118] To distinguish different pressurizing pieces **20**, the wrong pressurizing piece **20** is prevented from being assembled to the base **10**, thereby preventing the wrong pressurizing piece **20** from damaging the battery module. With the fool-proofing pieces **20***c* of different pressurizing pieces **20** designed to be different, it is possible to prevent the wrong pressurizing pieces **20** from being assembled to the base **10**, and ensure the accuracy of replacing the pressurizing pieces **20** while automatically switching between different models.

[0119] It should be noted that the fool-proofing pieces **20***c* of different pressurizing pieces **20** are different, for example, the sizes of the fool-proofing pieces **20***c* of different pressurizing pieces **20** are different, or the shapes of the fool-proofing pieces **20***c* of different pressurizing pieces **20** are different.

[0120] The fastener **12** is provided with a fool-proofing groove at the top, and the pressurizing piece **20** is provided with a fool-proofing piece **20***c* in one-to-one correspondence with the fool-proofing groove at the top, so that the accuracy of replacing the pressurizing piece **20** can be ensured, and the wrong pressurizing piece **20** can be prevented from damaging the battery module. In addition, the fool-proofing piece **20***c* is fitted to the fool-proofing groove, which can further position the pressurizing piece **20** and the base **10**.

[0121] In some embodiments, referring to FIG. **5**, a width of the base **10** is L**1**, where 100 mm≤L**1**≤1200 mm.

[0122] For example, the width L**1** of the base **10** can be measured by a vernier caliper at a temperature of 25° C.

[0123] For example, the width L**1** of the base **10** may be 100 mm, 105 mm, 120 mm, 125 mm, 130 mm, 149 mm, 171 mm, 200 mm, 218 mm, 279 mm, 301 mm, 340 mm, 341 mm, 395 mm, 412 mm, 498 mm, 520 mm, 585 mm, 606 mm, 666 mm, 702 mm, 803 mm, 926 mm, 990 mm, 1000 mm, 1010 mm, 1122 mm, 1130 mm, 1117 mm, 1200 mm, or the like.

[0124] The width L**1** of the base **10** is 100 mm≤L**1**≤1200 mm, achieving compatibility with more types of pressurizing pieces **20**, thereby improving the versatility of the base **10**. For example, if the width of the base **10** is set to be large enough, it can be compatible with more pressurizing pieces **20**, it can be compatible with more single-row or double-row pressurizing pieces **20** of more projects, and it can make the structure of the pressurizing apparatus **100** or the silo apparatus **200** compact.

[0125] In some embodiments, referring to FIG. **5**, a width of the base **10** is L**1**, where 150 mm≤L**1**≤500 mm.

[0126] For example, the width L**1** of the base **10** may be 150 mm, 170 mm, 201 mm, 215 mm, 250 mm, 280 mm, 298 mm, 300 mm, 315 mm, 322 mm, 338 mm, 340 mm, 355 mm, 360 mm, 366 mm, 382 mm, 400 mm, 426 mm, 430 mm, 440 mm, 450 mm, 462 mm, 470 mm, 487 mm, 500 mm, or the like.

[0127] The width L**1** of the base **10** is 150 mm≤L**1**≤500 mm, achieving compatibility with more types of pressurizing pieces **20**, thereby improving the versatility of the base **10**. For example, if the width of the base **10** is set to be large enough, it can be compatible with more pressurizing pieces **20** and more single-row or double-row pressurizing pieces **20**, and it can make the structure of the pressurizing apparatus **100** or the silo apparatus **200** more compact.

[0128] According to some embodiments of this disclosure, referring to FIGS. **1** and **2**, this disclosure further provides a pressurizing apparatus **100**. The pressurizing apparatus **100** includes a tray and the pressurizing mechanism disclosed in any one of the above embodiments. The pressurizing mechanism is arranged on the tray. The base **10** is configured to carry the pressurizing piece **20**. The pressurizing piece **20** can be configured to pressurize the battery module.

[0129] In an embodiment of this disclosure, the pressurizing apparatus **100** is configured to pressurize the battery module.

[0130] The pressurizing apparatus **100** may further include a driving mechanism configured to drive the pressurizing piece **20** to reciprocate, thereby exerting pressure on the battery module.

[0131] One of the pressurizing piece **20** and the base **10** is provided with the positioning key **10***a*, and the other one is provided with the positioning groove **20***a*, so that the pressurizing piece **20** and the base **10** can be positioned through the fitting of the positioning key **10***a* and the positioning groove **20***a*. The installation structure is simple, the operation is convenient, and the connection and disconnection between the pressurizing piece **20** and the base **10** are facilitated, thereby alleviating the problems of complicated replacement operation and low replacement efficiency of the pressurizing piece **20**. In addition, the pressurizing piece **20** and the base **10** are positioned through the fitting of the positioning key **10***a* and the positioning groove **20***a*, which is conducive to automation to some extent. Further, different types of pressurizing pieces **20** can be provided with positioning keys **10***a* or positioning grooves **20***a* adapted to the base **10**, so that different types of pressurizing pieces **20** can share the base **10**, thereby reducing the cost and improving the production efficiency.

[0132] According to some embodiments of this disclosure, referring to FIGS. **1** and **2**, this disclosure further provides a silo apparatus **200**. The silo apparatus **200** includes a case **210** and the pressurizing mechanism disclosed in any one of the above embodiments. The case **210** is provided with a feeding area **210***a* and an unloading area **210***b*. The feeding area **210***a* and the unloading area **210***b* are both provided with the pressurizing mechanism. The base **10** is configured to carry the pressurizing piece **20**.

[0133] The feeding area **210***a* configured to carry the pressurizing piece **20** to be used.

[0134] The unloading area **210***b* configured to carry the pressurizing piece **20** to be replaced.

[0135] The silo apparatus **200** is provided with the feeding area **210***a* and the unloading area **210***b*, thereby alleviating the problems of complicated replacement operation and low replacement efficiency of the pressurizing piece **20**.

[0136] One of the pressurizing piece **20** and the base **10** is provided with the positioning key **10***a*, and the other one is provided with the positioning groove **20***a*, so that the pressurizing piece **20** and the base **10** can be positioned through the fitting of the positioning key **10***a* and the positioning groove **20***a*. The installation structure is simple, the operation is convenient, and the connection and disconnection between the pressurizing piece **20** and the base **10** are facilitated, thereby alleviating the problems of complicated replacement operation and low replacement efficiency of the pressurizing piece **20**. In addition, the pressurizing piece **20** and the base **10** are positioned through the fitting of the positioning key **10***a* and the positioning groove **20***a*, which is conducive to automation to some extent. Further, different types of pressurizing pieces **20** can be provided with positioning keys **10***a* or positioning grooves **20***a* adapted to the base **10**, so that different types of pressurizing pieces **20** can share the base **10**, thereby reducing the cost and improving the production efficiency.

[0137] In some embodiments, still referring to FIGS. **1** and **2**, the feeding area **210***a* includes a plurality of feeding sub-areas arranged along the height direction, and each feeding sub-area is provided with a plurality of pressurizing mechanisms.

[0138] The feeding area **210***a* includes a plurality of feeding sub-areas arranged along the height direction, so that the space of the case **210** can be fully utilized, the structure is compact, and more pressurizing mechanisms can be placed in the feeding area **210***a*, improving the production efficiency.

[0139] In some embodiments, still referring to FIGS. **1** and **2**, the unloading area **210***b* includes a plurality of unloading sub-areas arranged along the height direction, and each of the unloading sub-areas is provided with a plurality of pressurizing mechanisms.

[0140] The unloading area **210***b* includes a plurality of unloading sub-areas arranged along the height direction, so that the space of the case **210** can be fully utilized, the structure is compact, and more pressurizing mechanisms can be placed in the unloading area **210***b*, improving the production efficiency.

[0141] In some embodiments, still referring to FIGS. **1** and **2**, the silo apparatus **200** includes a storage vehicle **220** configured to store the pressurizing mechanisms, and the pressurizing pieces **20** to be used are mounted on the base **10** of the pressurizing mechanism.

[0142] According to some embodiments of this disclosure, still referring to FIGS. **1** and **2**, this disclosure further provides a battery production line **1000** including a manipulator **300**, the pressurizing apparatus **100** disclosed in any one of the above embodiments, and the silo apparatus **200** in any one of the above embodiments. The manipulator **300** is configured to transfer and assemble the pressurizing piece **20** in the feeding area **210***a* to the base **10** of the pressurizing apparatus **100**, and transfer and assemble the to-be-replaced pressurizing piece **20** in the pressurizing apparatus **100** to the base **10** in the unloading area **210***b*.

[0143] For example, the manipulator **300** is, for example, a four-axis robot.

[0144] A process of replacing the pressurizing piece **20** may include the following steps: [0145] gripping the to-be-replaced pressurizing piece **20** on the pressurizing apparatus **100** by the manipulator **300**, and moving to release the connection between the base **10** and the to-be-replaced pressurizing piece **20** of the pressurizing apparatus **100**; [0146] mounting the to-be-replaced pressurizing piece **20** on the base **10** of the unloading area **210***b* of the silo apparatus **200**, for the pressure piece **20** to be replaced to go offline; [0147] according to the characteristics (size, shape, or the like) of the fool-proofing groove at the top of the fastener **12**, determining the characteristics (size, shape, or the like) of the fool-proofing groove at the top of the to-be-used pressurizing piece **20**, and determining the gripping sequence of the to-be-used pressurizing piece **20**; [0148] gripping the to-be-used pressurizing piece **20** in the feeding area **210***a* of the silo apparatus **200** by the manipulator **300**, and moving to release the connection between the base **10** and the to-be-used pressurizing piece **20** in the feeding area **210***a* of the silo apparatus **200**; [0149] mounting the to-be-used pressurizing piece **20** on the base **10** of the pressurizing apparatus **100**, to install the to-be-used pressurizing piece **20**; and [0150] leaving the base **10** of the pressurizing apparatus **100** flowing into the next station, and the next base **10** flowing into this station, thereby completing the action.

[0151] It should be noted that the order of the above steps does not need to be in strict accordance with the order disclosed above. For example, the to-be-used pressurizing piece **20** in the feeding area **210***a* of the silo apparatus **200** is gripped first by the manipulator **300**, and then the to-be-replaced pressurizing piece **20** on the pressurizing apparatus **100** is gripped by the manipulator **300**, as long as the pressurizing piece **20** can be replaced.

[0152] The movement of the manipulator **300** is electrically controlled. Through the pressurizing mechanism and electrical control, the pressurizing piece **20** can be automatically replaced, reducing the labor cost, improving the replacement efficiency and production capacity of the production line, and realizing replacement of the pressurizing piece **20** without shutting down the machine.
[0153] The to-be-used pressurizing piece **20** can be assembled to the base **10** manually, or the to-be-used pressurizing piece **20** can be assembled to the base **10** by the manipulator **300**.
[0154] The manipulator **300** can grip the two ends of the top of the pressurizing piece **20** or the grip portion **20***b* of the pressurizing piece **20**.
[0155] In a specific embodiment, referring to FIGS. **1** to **6**, the base **10** is provided with a plurality of positioning keys **10***a*, and the pressurizing piece **20** is provided with positioning grooves **20***a*. The base **10** is provided with an avoidance groove **10***b*, and the avoidance groove **10***b* is configured to avoid the pressurizing piece **20**. The pressurizing piece **20** includes a connecting portion **22** and a pressurizing portion **21** connected to the connecting portion **22**. The pressurizing portion **21** is provided with a positioning groove **20***a*. The positioning groove **20***a* is provided with an insertion opening penetrating the bottom wall of the pressurizing piece **20**. The positioning key **10***a* is inserted into the positioning groove **20***a* through the insertion opening. The pressurizing piece **20** is provided with a grip portion **20***b* at the top. The grip portion **20***b* is configured to cooperate with the manipulator **300** in gripping. The base **10** includes a base body **11** and a fastener **12** arranged on the base body **11**. The positioning key **10***a* is provided on a side surface of the fastener **12**. The pressurizing piece **20** includes a connecting body **211** and a limiting piece **212**. The connecting body **211** is provided with a groove. The limiting piece **212** is arranged at the edge of the groove and defines a positioning groove **20***a* with the groove. The positioning key **10***a* is arranged on the side surface of the fastener **12**, and a limiting groove **10***c* is formed between the positioning key **10***a* and the fastener **12**. When the positioning key **10***a* is fitted to the positioning groove **20***a*, the limiting piece **212** is snapped into the limiting groove **10***c* to limit the pressurizing piece **20** and the base **10**. The fastener **12** is provided with a fool-proofing groove at the top. The pressurizing piece **20** is provided with a fool-proofing piece **20***c* corresponding one-to-one with the fool-proofing groove at the top. When the positioning key **10***a* is fitted to the positioning groove **20***a*, the fool-proofing piece **20***c* is snapped into the fool-proofing groove. The fool-proofing pieces **20***c* of different pressurizing pieces **20** are different.
[0156] The foregoing embodiments are merely intended to describe the technical solutions of this disclosure but not to limit this disclosure. Although this disclosure has been described in detail with reference to the foregoing embodiments, those of ordinary skill in the art understand that modifications may still be made to the technical solutions described in the foregoing embodiments, or equivalent replacements may still be made to some or all technical features in the technical solutions. Such modifications and replacements do not make the essence of the corresponding technical solutions depart from the scope of the technical solutions of the embodiments of this disclosure, but still fall within the scope of this disclosure. In particular, the various technical features mentioned in the various embodiments may be combined in any manner as long as there is no structural conflict. This disclosure is not limited to the specific embodiments disclosed herein, but includes all technical solutions falling within the scope of this disclosure.
Industrial Practicality
[0157] In this disclosure, the movement of the manipulator is electrically controlled. Through the pressurizing mechanism and electrical control, the pressurizing piece can be automatically replaced, reducing the labor cost, improving the replacement efficiency and the production capacity of the production line, and realizing replacement of the pressurizing piece without shutting down the machine.

Claims

1. A pressurizing mechanism, wherein the pressurizing mechanism comprises: a base; and a pressurizing piece arranged on the base, wherein the base is provided with a positioning key, the pressurizing piece is provided with a positioning groove, and the positioning key is fitted to the positioning groove to position the pressurizing piece and the base; and the positioning groove is provided with an insertion opening running through a bottom wall of the pressurizing piece, and the positioning key is inserted into the positioning groove through the insertion opening.
2. The pressurizing mechanism according to claim 1, wherein the base is provided with an avoidance groove, and the avoidance groove is configured to avoid the pressurizing piece.
3. The pressurizing mechanism according to claim 1, wherein the pressurizing piece comprises a connecting portion and a pressurizing portion connected to the connecting portion, and the pressurizing portion is provided with the positioning groove.
4. The pressurizing mechanism according to claim 3, wherein the pressurizing portion comprises a connecting body and a limiting piece, wherein the connecting body is provided with a groove, and the limiting piece is arranged at an edge of the groove and defines the positioning groove with the groove.
5. The pressurizing mechanism according to claim 1, wherein the pressurizing piece is provided with a grip portion at the top, and the grip portion is configured to cooperate with a manipulator in gripping.
6. The pressurizing mechanism according to claim 1, wherein the base is provided with a plurality of positioning keys.
7. The pressurizing mechanism according to claim 1, wherein the base comprises a base body and a fastener arranged on the base body, and the positioning key is arranged on a side surface of the fastener.
8. The pressurizing mechanism according to claim 7, wherein the pressurizing piece comprises a connecting body and a limiting piece, wherein the connecting body is provided with a groove, and the limiting piece is arranged at an edge of the groove and defines the positioning groove with the groove; and the positioning key is arranged on a side surface of the fastener, and a limiting groove is formed between the positioning key and the fastener; and when the positioning key is fitted to the positioning groove, the limiting piece is snapped into the limiting groove to limit the pressurizing piece and the base.
9. The pressurizing mechanism according to claim 7, wherein the fastener is provided with a fool-proofing groove at the top, the pressurizing piece is provided with a fool-proofing piece in one-to-one correspondence with the fool-proofing groove at the top, and when the positioning key is fitted to the positioning groove, the fool-proofing piece is snapped into the fool-proofing groove, wherein dimensions of the fool-proofing pieces of different pressurizing pieces are different.
10. The pressurizing mechanism according to claim 1, wherein a width of the base is L**1**, wherein 100 mm≤L**1**≤1200 mm.
11. The pressurizing mechanism according to claim 1, wherein a width of the base is L**1**, wherein 150 mm≤L**1**≤500 mm.
12. A pressurizing apparatus, comprising a tray and the pressurizing mechanism according to claim 1, wherein the pressurizing mechanism is arranged on the tray, and the pressurizing mechanism is configured to pressurize a battery module on the tray.
13. A silo apparatus, comprising: a case, wherein the case comprises a feeding area and an unloading area; and the pressurizing mechanism according to claim 1, wherein the feeding area and the unloading area are both provided with the pressurizing mechanism.
14. The silo apparatus according to claim 13, wherein the feeding area comprises a plurality of feeding sub-areas arranged along a height direction, and each of the feeding sub-areas is provided with a plurality of the pressurizing mechanisms; and/or the unloading area comprises a plurality of unloading sub-areas arranged along the height direction, and each of the unloading sub-areas is

provided with a plurality of the pressurizing mechanisms.

15. A battery production line, comprising: a pressurizing apparatus, comprising a tray and the pressurizing mechanism according to claim 1, wherein the pressurizing mechanism is arranged on the tray, and the pressurizing mechanism is configured to pressurize a battery module on the tray; a silo apparatus, comprising: a case, wherein the case comprises a feeding area and an unloading area; and the pressurizing mechanism; wherein the feeding area and the unloading area are both provided with the pressurizing mechanism; and a manipulator, wherein the manipulator is configured to transfer and assemble the pressurizing piece in the feeding area to the base of the pressurizing apparatus, and transfer and assemble the to-be-replaced pressurizing piece in the pressurizing apparatus to the base in the unloading area.